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Somaiya Vidyavihar University
K J Somaiya School of Engineering
Department of Information Technology

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Urine Test Strip Analysis Using Colorimetry and DevOps Microservices

Course : DevOps Lab Mini Project Poster Presentation

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Class: Information Technology | Div: ___ | Batch: ___

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1. INTRODUCTION

- **Clinical Reagent Strips:** Multistix / 10-SG urine test strips provide point-of-care screening for 10 key biochemical parameters (Glucose, Protein, pH, Ketones, etc.).
- **Human Subjectivity Challenge:** Traditional visual inspection against printed color charts is compromised by ambient lighting variations, visual perception bias, and delayed readings.
- **Engineering Solution:** An automated, containerized microservices platform combining computer vision, quantitative CIE Lab colorimetry, and Machine Learning.

2. PROBLEM STATEMENT

- **Problem:** Manual color comparison across 10 reagent pads simultaneously is error-prone, lacks audit trails, and suffers from colorimetric inaccuracy under non-standard light sources.
- **Architectural Dilemma:** Monolithic health apps lack fault resilience; a single ML or vision failure crashes the entire clinical dashboard.

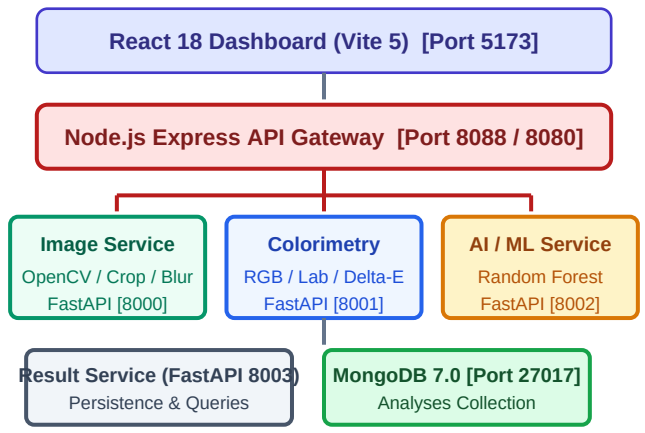
3. OBJECTIVES

- **Computer Vision Pipeline:** Detect urine strips, validate image quality (Laplacian blur & brightness thresholds), and extract precise pad coordinates.
- **Perceptual Colorimetry:** Sample central median pad pixels to eliminate edge glare; convert RGB to HSV and CIE Lab for Delta-E distance matching.
- **Explainable AI Assist:** Train an offline Random Forest classifier on synthetic reference swatches to output estimated categories and confidence scores.
- **Microservice Architecture:** Decouple image processing, colorimetry, AI, and storage into independent FastAPI containers behind an Express Gateway.

4. LITERATURE REVIEW

- **Shen et al. (2012):** Developed smartphone-based colorimetric detection for biochemical test strips; proved feasibility but highlighted sensitivity to variable lighting conditions.
- **Yetisen et al. (2014):** Evaluated color spaces for medical diagnostics; demonstrated that perceptual CIE Lab color space drastically minimizes visual variance compared to RGB.
- **Jalal et al. (2017):** Built mobile paper-based diagnostic readers; recommended central crop median filtering to prevent pad border reagent leakage artifacts.
- **Newman (2021) - Microservices:** Established architectural principles for distributed systems; recommended API gateways for request aggregation and circuit breaking.
- **Research Gap Identified:** Existing point-of-care projects are tightly-coupled monoliths lacking enterprise DevOps workflows, containerized isolation, and automated failure fallback mechanisms.

5. METHODOLOGY / SYSTEM DESIGN



DevOps CI/CD Automation Pipeline (Jenkins & GitHub Actions):

1. Env Setup -> 2. Train ML -> 3. Python Tests -> 4. Gateway Tests ->
5. Frontend Build -> 6. Docker Compose Build -> 7. Stack Health Deploy

Status: 100% Automated | All 7 Stages Passed (Build #6 SUCCESS)

6. IMPLEMENTATION DETAILS

- **Frontend Dashboard:** React 18 with Vite 5, interactive pad overlay boxes, real-time RGB/HSV/Lab colorimetric tables, and MongoDB historical query cards.
- **API Gateway:** Node.js Express server orchestrating multipart image forwarding, service aggregation, and graceful error recovery.
- **Image Service (CV):** OpenCV-based Laplacian variance calculation (rejects blur < 80); brightness check (35-245); bounding box generation for 10 vertical pads.
- **Colorimetry Service:** Extracts central 50% median color; converts RGB -> HSV -> CIE Lab; computes Euclidean Delta-E distance to reference chart colors.
- **AI/ML Service:** Scikit-Learn Random Forest (100 trees, 6 features [R,G,B,L,a,b]) loaded from joblib; outputs class probabilities.

7. RESULTS & DISCUSSION

Test & Build Validation Summary:

Component / Test Suite	Test Tool	Outcome
Python Microservices	Pytest 8.3	**14 / 14 Passed (100%)**
Node.js API Gateway	Jest 29	**3 / 3 Passed (100%)**
React Frontend Dashboard	Vitest 2.1	**1 / 1 Passed (100%)**
Production Bundle Build	Vite 5	**Compiled (147 kB)**
Multi-Container Build	Docker Compose	**6 / 6 Images Built**
Jenkins Automated Pipeline	Jenkins 2.568	**7 / 7 Stages Green (5m 57s)**
Fault Fallback Simulation	Stop AI Container	**Graceful Degradation Verified**

Demonstrated Microservices Fault Resilience:

- Executed 'docker compose stop ai-service' during live strip analysis.
- API Gateway automatically fell back to reference color matching.
- User receives analysis + warning; zero crashes; MongoDB retains record.

8. KEY FINDINGS

- **CIE Lab Perceptual Superiority:** Euclidean distance in CIE Lab space correctly identified demonstration color swatches with minimal error (Delta-E < 3.3).
- **Median Crop Noise Immunity:** 50% central median cropping successfully filtered out border shadows, reflections, and pad margin bleeding.
- **Fault Resilience Confirmed:** Disabling the AI service triggered instant gateway fallback to reference-matching with 0% downtime and explicit UI warnings.

9. CONCLUSION

- **Complete DevOps Solution:** Engineered an end-to-end multi-language microservices application successfully bridging Computer Vision, Colorimetry, and DevOps.
- **High Reliability & Maintainability:** Independent Docker containers ensure individual services can be scaled, updated, or restarted without impacting others.

10. FUTURE SCOPE

- **Standardized Color Calibration:** Integrate 24-color Macbeth/ColorChecker target boards in strip photos for automatic white-balance correction.
- **Multi-Brand Layout Recognition:** Use Deep Learning (YOLOv8) for dynamic multi-manufacturer strip detection (Siemens, Roche, Mission).
- **Time-Resolved Kinetic Capture:** Implement mobile camera timer alerts to measure specific reagent pad reactions at 30s, 60s, and 120s intervals.

11. REFERENCES

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